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APPENDIX A. THE PROOF OF LEMMA 2.6

In this section, we will give a generalized version of Wu’s algorithm that can be applied to RS codes defined on a non-cyclic multipoint set. Let us first recap some notations in syndrome decoding.

Assume $\mathcal{P} = \{\xi, \xi\alpha, \dots, \xi\alpha^{n-1}\} \in \mathbb{F}_q^*/\langle\alpha\rangle$ is a coset of cyclic group, where $\alpha \in \mathbb{F}_q^*$ has order n . The RS $[n, k]$ that we concerned is defined as

$$\text{RS}[n, k] = \{(f(\xi), f(\xi\alpha), \dots, f(\xi\alpha^{n-1})) \mid f(x) \in \mathbb{F}_q[x]_{<k}\}.$$

Let $r = n - k$. For a burst interval $[i, i + r - 2]$, we always view it as a subset of $[1, n]$ in the following sense

$$[i, i + r - 2](\bmod n + 1) = \begin{cases} [i, i + r - 2], & \text{if } i + r - 2 \leq n; \\ [i, n] \cup [1, (i + r - 2 \bmod n + 1) + 1], & \text{if } i + r - 2 > n. \end{cases}$$